

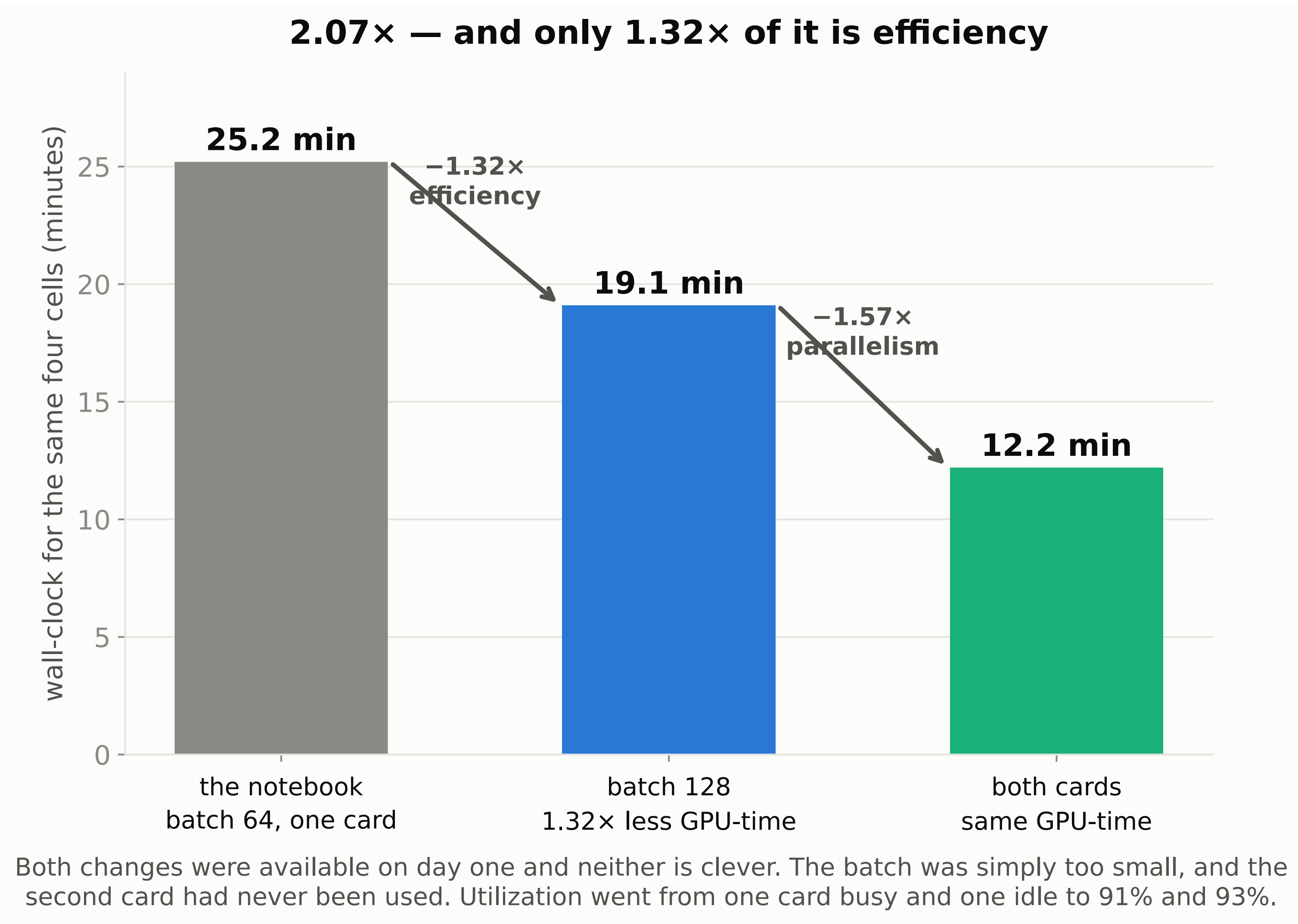
Building a Model Factory

From one-off notebooks to repeatable, challengeable experiments

W

TAKEAWAY The factory made hundreds of model comparisons affordable and auditable—then gave collaborators enough control to disprove results that initially looked convincing.

MEASURE



From notebook state to evidence

A scaling study varies data, compute, model size, language, seed and learning rate. Managing that grid by notebook state and filenames is not reproducibility. The factory turns each cell into a self-describing job, checkpoint and result record.

The same four cells fell from 25.2 to 12.2 minutes: 2.07× end-to-end. Only 1.32× was true efficiency from a better batch; the rest was two cards working in parallel. Reporting the decomposition keeps hardware scale from masquerading as algorithmic efficiency.

9

public MLM API functions

2.07×

faster end-to-end · 1.32× efficiency

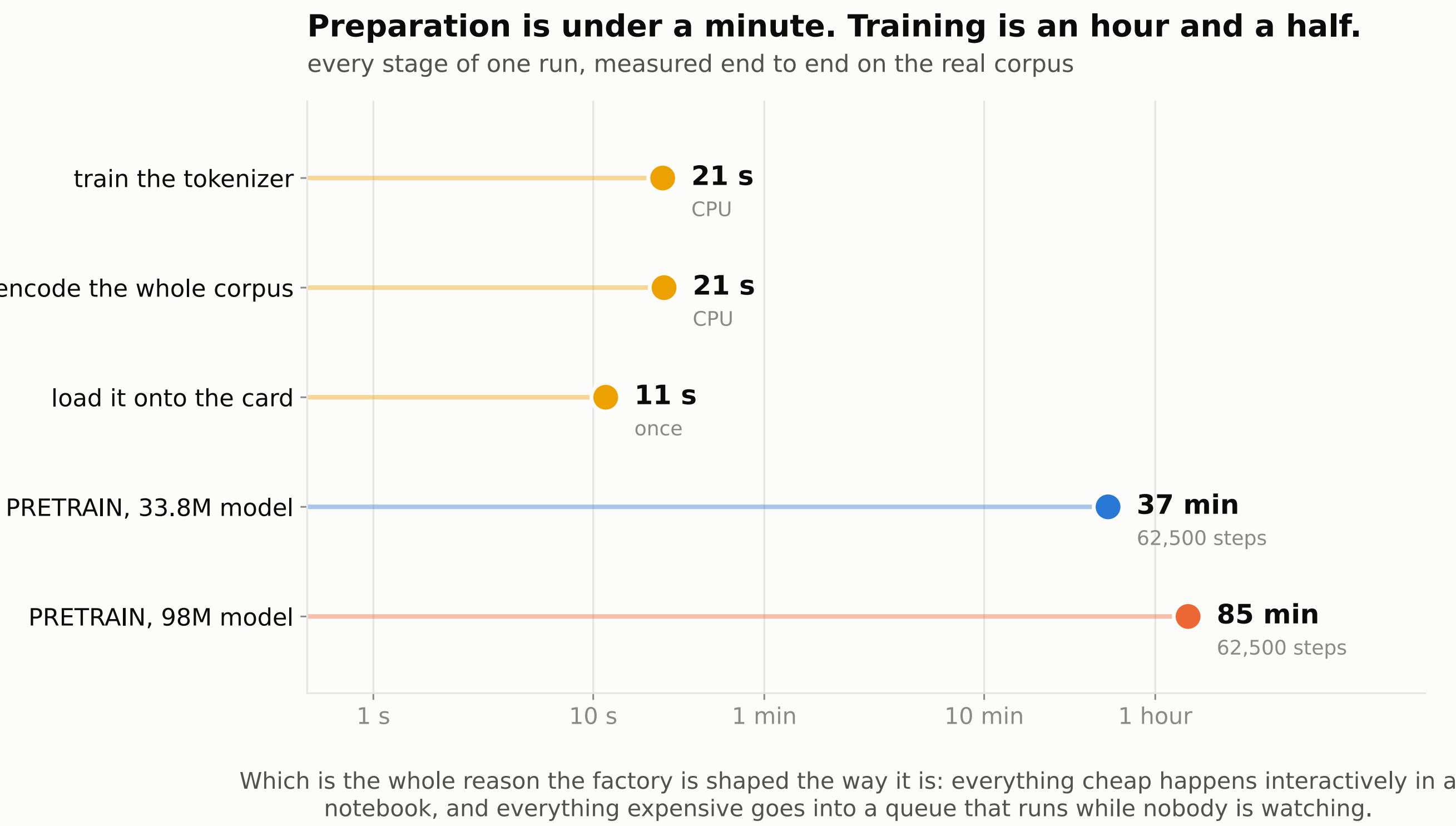
197

pretraining run records

1.024B

update tokens in a full budget

1 · SPLIT BY COST



2 · HIDE COMPLEXITY



3 · LIMIT THE CLAIM

Three seeds tells you what is definitely noise, not what is definitely real

